

Minji Shon

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Summary

Ph.D. researcher in ECE (Georgia Tech) with **7 years of Samsung Foundry Design-for-Reliability** experience, specializing in **reliability- and thermal-aware DTCO/STCO** for advanced 3D CMOS (CFET, GAA nanosheet) — spanning TCAD device modeling, RTL-to-GDS implementation, and reliability sign-off.

Education

Georgia Institute of Technology  — Atlanta, GA, USA Aug 2023–Present

Ph.D., Electrical and Computer Engineering (*Expected May 2028*)

Sogang University  — Seoul, South Korea Mar 2012–Aug 2016

B.S., Electronic Engineering; **summa cum laude**

University of Connecticut  — Storrs, CT, USA Jan–May 2015

International Student Exchange, Electrical and Computer Engineering

Professional Experience

Graduate Research Assistant — Georgia Institute of Technology  , GA, USA Aug 2023–Present

SRC Research Scholar (SPARC & PRISM & CHIMES)

Ph.D. thesis: *Next-Generation 3D Scaling: Reliability-Aware Device Modeling and Design Co-Optimization for Advanced CMOS Logic and Memory* (Advisor: Dr. Shimeng Yu — [Laboratory for Emerging Devices and Circuits](#)).

- **Open CFET PDK, multi-tier RTL-to-GDS flow & logic DTCO** — TCAD-calibrated device models and multi-tier standard-cell libraries enabling academic full-stack DTCO; cell- to block-level PPA and routing-congestion benchmarking (RISC-V core, systolic array) on commercial (Cadence/Synopsys) tools. [[L1](#), [L2](#), [L3](#)]
- **Reliability-aware co-optimization** — device-to-system electro-thermal framework; GPU-accelerated multi-physics (DeepSim) extended to multi-tier CFET. [[R1](#), [R2](#)]
- **Broader contributions across the memory pyramid** — CFET DCIM (memory wall) and ferroelectric-NAND gate-stack modeling (beyond vertical NAND). [[L4](#), [F1–F4](#)]

Design Quality & Reliability Engineer Intern — Intel  , Hillsboro, OR, USA Jun 2026–Jan 2027

- Product Quality Network team: developing Tools, Flows, and Methods (TFM) for pre-silicon quality and reliability validation of advanced CPU, Server, and ASIC products.

Technology Quality & Reliability Engineer — Samsung Electronics **SAMSUNG**, Korea Aug 2016–Jul 2023

Implemented aging-aware circuit design environment

- Developed aging-aware circuit design environments and PDK components for **3 nm GAA nanosheet** technology.

Enhanced reliability model-to-hardware correlation (MHC) & DFR environment

- Built/validated reliability models for **HCI, BTI, TDDb, self-heating**, improving prediction accuracy using **ring oscillator** benchmarks in collaboration with WLR and PDK teams.
- Developed **statistical aging simulation modeling and methodology**, enabling **AEC-Q100** automotive-grade DFR in **8 nm FinFET** products.

Advanced IP & product-level reliability verification

- Delivered aging/variability-aware design methodologies for **Samsung Exynos SoCs** (14 nm–4 nm) under advanced reliability sign-off.
- Provided DFR for real operating conditions (multi-step aging, power-down bias instability) and verified **overdrive-voltage reliability** in NVM (eFuse, OTP, MRAM). [[R3](#)]
- Collaborated with Design Technology Enablement to provide **critical-path reliability timing margin** via aging simulation for Exynos SoCs.

Technology Quality & Reliability Intern — Samsung Electronics **SAMSUNG**, Korea Jul–Aug 2016

- Implemented **enhanced Dynamic Device Voltage Check (DDVC)**; strengthened **safe-operating-area (SOA)** validation and improved **voltage-stress reliability sign-off** across advanced nodes.

Technical Skills

Focus: reliability-/thermal-aware DTCO/STCO; multi-tier CFET & 3D integration; ferroelectric device modeling; aging/reliability sign-off.

TCAD & device: Sentaurus electro-thermal device/process; BSIM-CMG compact modeling.

RTL-to-GDS & sign-off flows: Cadence & Synopsys synthesis, P&R and reliability sign-off; full PDK build & verification (LIB/LEF, LPE, DRC/LVS).

Circuit & reliability simulation: SPICE (HSPICE/FineSim/Spectre) with aging engines (RelXpert/MOSRA).

Programming & HDL: Python, TCL, Perl, C, MATLAB, SystemVerilog.

Publications (Selected, by Topic)

* denotes equal contribution.

3D Logic & SRAM DTCO/STCO

- [L1] **Shon, M.***, Lim, S.*, et al., Yu, S. (2026). *MATRIX-A5: An Open-Source CFET PDK with CfetFlex Track-Mixing Libraries for RTL-to-GDS DTCO*. JXCDC (in preparation).
- [L2] **Shon, M.***, Lim, S.*, Yu, S. (2025). *DTCO of Multi-Tier CFETs: Ultimate Logic and SRAM Scaling of A5 Technology and Beyond*. IEDM'25. [DOI](#)
- [L3] Lim, S., **Shon, M.**, Yu, S. (2025). *3-Tier SRAM Complementary FET (CFET) High Density (HD) Cell Design at A5 Node*. TED'25. [DOI](#)
- [L4] Lee, J.*, **Shon, M.***, et al., Yu, S. (2025). *3D Digital Compute-in-Memory Benchmark with A5 CFET Technology: An Extension to Look-up-Table Based Design*. TVLSI'25. [DOI](#) (earlier version: APCCAS'24, [DOI](#))

Ferroelectric Device & Applications

- [F1] Fernandes, L.*, **Shon, M.***, et al., Khan, A. (2026). *Channel-Side Interlayer Engineering in Laminated FeFETs: Trap-Driven Optimization of Memory Window, Retention, and Disturb in Ferroelectric NAND Flash*. IRPS'26. [DOI](#)
- [F2] Phadke, O., **Shon, M.**, et al., Yu, S. (2025). *On the Localized Ferroelectric Phase Variation in Scaled FeFET: Experiments and Modeling*. IEDM'25. [DOI](#)
- [F3] **Shon, M.**, et al., Yu, S. (2025). *Modeling Dynamic Interplay Between Charge Traps and Polarization for Memory Window Enhancement in Gate Injection Layers*. IRPS'25. [DOI](#)
- [F4] **Shon, M.**, et al., Yu, S. (2025). *A Comprehensive Modeling of Gate Stack Interlayer Engineering for Ferroelectric Vertical NAND*. DRC'25. [DOI](#)

Reliability & Electro-Thermal Co-Design

- [R1] **Shon, M.***, McClellan, C. J.*, et al., Yu, S. (2026). *AI-Accelerated and Layout-Resolved Electro-Thermal Analysis for Multi-Tier CFET*. IEDM'26 (in preparation).
- [R2] Dutta, R.*, **Shon, M.***, et al., Datta, S. (2026). *Self-Heating in GAA Nanosheet FETs with integrated Back Side Power Delivery Network: Device to Circuit Electro-Thermal Analysis and Implications*. IRPS'26. [DOI](#)
- [R3] Shim, H., Jo, J., Kim, Y., Jeong, B., **Shon, M.**, et al., Pae, S. (2019). *Aging-Aware Design Verification Methods Under Real Product Operating Conditions*. IRPS'19. [DOI](#)

Honors & Awards

- **Best Paper Award, Runner-Up** — IEEE Device Research Conference (DRC) Jun 2025
- **Silicon Creations ISSCC Travel Grant** Feb 2024
- **Conference Travel Grants** — APCCAS'24, IRPS, DRC, IEDM'25 (presentation) 2024–2025
- **CSE Department Fellowship (offered)** — University of Minnesota–Twin Cities Jan 2023
- **Summa cum laude** — Sogang University; Dean's List Aug 2016
- **Full National Scholarship (Merit)** — Korea Student Aid Foundation 2014–2016
- **Honors Scholarship (Merit)** — Sogang University 2012–2015

Leadership & Service

- **Ministry Team Leader, Grad CRU** — Georgia Institute of Technology 2024–Present
- **Education Mentor / Sub-Course Manager** — Samsung Electronics 2019,2020,2022
 - Delivered onboarding for 200+ new hires; “Global Business Manners,” “Communication & Creative Thinking.”